Programa2.Estandarizacion

November 7, 2026

0.1 Programa2.Estandarización

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0.1.1 2 Estandarización

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Link Repositorio: https://github.com/eulloaitq/Fundamentos-de-Machine-Learning.git

[3]:
```
from sklearn import preprocessing
import numpy as np
```

[4]:
```
# Carga de datos.
X_train = np.array([[ 1., -1., 2.],
[ 2., 0., 0.],
[ 0., 1., -1.]])
```

[5]:
```
# Limpieza de datos: estandarización.
standardizer = preprocessing.StandardScaler()
# Estandarizamos los datos de entrenamiento
X_train_std=standardizer.fit_transform(X_train)
print(X_train_std)
```

```
[[ 0.         -1.22474487  1.33630621]
 [ 1.22474487  0.         -0.26726124]
 [-1.22474487  1.22474487 -1.06904497]]
```

[6]:
```
X_train_2 = np.array([[ 1., -1., 2.],
[ 2., 0., 0.],
[ 0., 1., -1.]])
```

[7]:
```
# Aplicamos los métodos "fit" y "transform" por separado
f=preprocessing.StandardScaler()
f=f.fit(X_train)
X_train_2_std = f.transform(X_train_2)
print(X_train_2_std)
# ???
```

```
[[ 0.         -1.22474487  1.33630621]
 [ 1.22474487  0.         -0.26726124]
 [-1.22474487  1.22474487 -1.06904497]]
```

[8]:
```
# Comprobar si las matrices estandarizadas son iguales
# ???
print(np.array_equiv(X_train_std, X_train_2_std))
```

```
True
```